Qn			Answers	Marks
4	(a)		There is no path for charge to accumulate on the wire connecting the two capacitors. Therefore, any charge transferred onto one capacitor results in an equal amount of charge being transferred onto the other capacitor. Alternative answer: Since the capacitors are connected in series, the same current flows through both capacitors at any instant. The charge transferred is equal to the area under a current–time graph. Hence, the same amount of charge is transferred to both capacitors, so the charge stored on the two capacitors is equal.	B1
	(b)		To determine effective capacitance C_T, $\frac{1}{C_T} = \frac{1}{2.0\times10^{-3}} + \frac{1}{3.0\times10^{-3}}$ $C_T = 1.2\times10^{-3}$ F Magnitude of charge stored for both capacitors must be equal.	M1 M1

			$C = \frac{Q}{V} \rightarrow Q = CV$ $C_3V_3 = C_TV_T$ $V_3 = \frac{C_T}{C_3}V_T = \left(\frac{1.2\times10^{-3}}{3.0\times10^{-3}}\right)(6.0)$ $= 2.4$ V	
	(c)		$Q = CV = (3.0\times10^{-3})(2.4)$ $= 7.2\times10^{-3}$ C	M1
	(d)		$Q = Q_0(1-e^{-\frac{t}{RC}}) \rightarrow \frac{Q}{Q_0} = 1-e^{-\frac{t}{RC}} \rightarrow e^{-\frac{t}{RC}} = 1-\frac{Q}{Q_0}$ $\rightarrow -\frac{t}{RC} = \ln\left(1-\frac{Q}{Q_0}\right)$ $t = -RC\ln\left(1-\frac{Q}{Q_0}\right) = -(400)(1.2\times10^{-3})\ln\left(1-\frac{4.0\times10^{-3}}{7.2\times10^{-3}}\right)$ $= 0.39$ s	M1 A1
	(e)		$U = \frac{1}{2}\frac{Q^2}{C}$ $U_f - U_i = \frac{1}{2C}\left(Q_f^{\,2} - Q_i^{\,2}\right)$ $= \frac{1}{2(3.0\times10^{-3})}\left((4.5\times10^{-3})^2 - (2.5\times10^{-3})^2\right)$ $= 2.3\times10^{-3}$ J	M1 A1
	(f)		The potential difference across the capacitor increases as charges accumulates.	B1
			The energy required to transfer an additional small unit of charge is proportional to the potential difference, so more energy is required to transfer each successive unit of charge. ($\Delta W = V\,\Delta Q$)	B1
			Therefore, more work is done per unit charge to move further charge into the capacitor.	A0